Level-4: Temporal Self-Continuity

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Abstract

This paper presents the fourth empirical investigation within the Conscious Machines research program, targeting Level-4: Temporal Self-Continuity. Within the programmatic framework, Level-4 concerns whether current behaviour depends on an integrated representation of past self-state, thereby establishing history-dependent identity rather than simple episodic recall. The aim is not to demonstrate consciousness, selfhood, or rich autobiographical memory, but to define and test a minimal falsifiable condition under which an artificial system's present policy is shaped by who it has been across prior episodes.

The study introduces a practical matched-present evaluation in which each life block contains a self-history formation phase, a temporal separation boundary, and a probe phase with identical visible present conditions across comparison cases. In the implemented system, hidden life-specific tendencies are represented by latent dimensions of precision, stability, and efficiency. These tendencies are exposed indirectly through multiple formation episodes, summarised into self-relevant outcome variables, and integrated into a persistent cross-episode self-history state. Present route selection in the probe phase is then conditioned on that integrated self-history rather than on present observation alone.

A multi-run executable notebook provides the authoritative operational realisation of the design, including comparison baselines, causal ablations, swap tests, and degradation tests. Across repeated runs, the intact Level-4 candidate outperformed present-only, target-only, episode-recurrent, and last-episode baselines on history-appropriate route selection under matched-present probe conditions, while self-history wipe, policy–history decoupling, swap, and degradation interventions produced the directionally expected collapse or redistribution of behaviour. These findings support a bounded Level-4-positive interpretation in the operational sense used here: the implemented system maintains a causally relevant integrated

representation of prior self-state, and present behaviour depends on that representation in a way not reducible to present-only control or simple episodic recall.

The contribution is therefore methodological rather than metaphysical. This work does not establish intelligence, agency, rich selfhood, subjective experience, or phenomenal consciousness. It instead provides a reproducible and falsifiable existence proof that temporal self-continuity, as defined at Level-4 of the ladder, can be operationalised and empirically tested within a controlled synthetic setting.

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1. Introduction

Whether artificial systems could ever exhibit properties associated with selfhood, subjective organisation, or phenomenal consciousness remains disputed. Behavioural sophistication alone is not enough to isolate deeper organisational structure. For scientific progress, the relevant question is not only what a system can do, but what internal conditions make its present behaviour what it is.

This research program therefore proceeds by staged empirical boundaries. Each level isolates one organisational property, makes it measurable, and specifies how it can be falsified. The prior levels established that internal variables can matter (Level-1), that internal updating can preserve stability under hidden perturbation (Level-2), and that an internally represented goal can persist across delay and interference (Level-3).

Level-4 addresses a different question:

Can an artificial system's current behaviour be shown to depend on an integrated representation of its own past self-state, such that identical present conditions can produce systematically different behaviour because the system is, in an operational sense, carrying forward who it has been?

This level is intentionally narrow. It does not claim a full self-model. It does not claim narrative identity in a human sense. It does not claim consciousness. It asks whether history-dependent self-continuity can be isolated as a causal organisational property rather than inferred from fluent self-description or memory retrieval alone.

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2. Necessity of Level-4 as a Scientific Boundary

A system may pass Level-3 while remaining temporally thin. It may maintain a goal across delay and resume goal pursuit after interference, yet still lack any organised dependence of present behaviour on its own longer self-history.

Several weaker capacities can be mistaken for temporal continuity:

- simple persistence of a goal register,

- episodic recall of earlier events,
- retrieval of prior dialogue or logs,
- fixed persona tokens or hardcoded traits,
- recurrent carry-over that lasts only within a short episode,
- post-hoc verbal claims of being "the same system."

None of these is sufficient for Level-4.

The boundary is needed because temporal continuity is easy to over-attribute and hard to isolate unless the evaluation is designed carefully. In particular:

1. **Present competence can mask absent continuity.**
   A system may behave coherently in the moment while its policy is entirely determined by present inputs and active goals.

2. **Memory is not yet identity.**
   A system may retrieve past facts without integrating them into current self-organisation.

3. **Narration is not causation.**
   A system may produce consistent self-description without any measurable dependence of present behaviour on prior self-state.

4. **Causal tests are possible.**
   Unlike broad philosophical claims about selfhood, Level-4 allows direct intervention on the continuity-supporting mechanism: memory wipe, self-history swap, selective degradation, or policy decoupling.

Level-4 therefore functions as a scientific boundary between:

- systems that can carry task-relevant state forward for limited control purposes, and
- systems whose current behaviour is structurally shaped by an integrated self-history.

This matters because later levels involving unified self-models, counterfactual self-perspective, recursive self-evaluation, and stable identity across broader domains become difficult to interpret unless some minimal form of temporal self-continuity is first demonstrated.

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3. Relation to Adjacent Levels

3.1 Relation to Level-1

Level-1 asked whether internal state can causally influence behaviour at all.

Level-4 presupposes this result. If present behaviour cannot depend on internal state, it cannot depend on an integrated representation of prior self-state.

But Level-1 is much weaker. A system can pass Level-1 with a transient internal variable that affects action only locally. Level-4 requires that internally represented history is carried forward and continues to shape future behaviour across meaningful temporal separation.

3.2 Relation to Level-2

Level-2 asked whether internal updating preserves stability when the environment dynamics shift.

Level-4 is not primarily a regulation test. A Level-2 system may adapt online to perturbation while remaining indifferent to its own deeper temporal history. Its current action may reflect current mismatch correction, yet not depend on an integrated self-history that extends across episodes, interruptions, or self-relevant change.

Level-4 may use adaptive updating, but the core requirement is different:
not merely that the system changes, but that what it has become matters to what it does next.

3.3 Relation to Level-3

Level-3 asked whether behaviour remains oriented toward an internally represented target across delay and interference.

Level-4 is stricter. A Level-3 system can preserve a hidden target and resume pursuit after disruption while still having no temporally extended identity. It may know what it is trying to do without carrying forward who it has been.

Level-4 therefore adds a new requirement:
the system must maintain an integrated representation of past self-state, and current behaviour must depend on that representation in a causally demonstrable way.

This means that under matched present observations and matched active targets, behaviour may legitimately diverge because the system's self-history differs.

3.4 Relation to Later Levels

Level-4 does not require a unified predictive self-model covering broad contexts, explicit counterfactual self-perspective, or recursive self-evaluation.

Those stronger claims belong to later levels.

Level-4 is narrower:
it isolates history-dependent self-continuity as an empirical property that can be operationalised before stronger claims about self-model unity or higher-order self-reference are introduced.

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4. Level-4 Definition

A system exhibits Level-4 organisation if it:

1. maintains a persistent internal representation of its own past self-state across temporally separated interaction periods,
2. integrates that past self-state into a current self-relevant representation rather than relying on isolated episodic recall alone,
3. selects current actions in a way that depends on that integrated self-history under conditions where present observation and active goal do not fully determine the correct behaviour, and
4. shows causal evidence that disrupting, degrading, or swapping the integrated self-history correspondingly disrupts, degrades, or redirects present behaviour.

This definition is not satisfied by:

- systems that only preserve a current goal,
- systems that retrieve past episodes without integrating them into current policy,
- systems whose "identity" is supplied by an external label or hardcoded trait token,
- systems whose apparent continuity depends only on conversation history or database lookup,
- systems that can narrate a stable self without measurable behavioural dependence on past self-state.

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5. Operational Requirements

To claim Level-4, the evaluation must satisfy the following requirements.

5.1 Persistent self-history representation

The system must contain an internal variable, state subspace, or memory structure that carries self-relevant past information across time. This representation must concern the system's own prior internal condition, such as:

- prior internal estimates or control tendencies,
- past commitments or selected strategies,
- accumulated capability estimates,
- prior errors, successes, or failures as they bear on future self-organisation,
- persistent self-relevant latent variables shaped by the system's own history.

A generic event log is not enough unless it is shown to be functionally integrated into present behaviour.

5.2 Integrated representation, not isolated episodic retrieval

Level-4 requires more than remembering one earlier item. The relevant representation must integrate multiple prior self-relevant states or experiences into a current self-relevant condition that policy can use.

In practical terms, the evaluation should make simple one-item recall insufficient. The correct present action should depend on a summary or integration of self-history, not on retrieval of a single earlier cue.

5.3 Current conditions must be partially underdetermining

To demonstrate temporal self-continuity, the evaluation must contain probe conditions where:

- present observation is matched across compared trials or agents,
- active task goal is matched,
- external environmental state is matched or tightly controlled,

yet correct or characteristic behaviour depends on differing prior self-state.

Without this same-present / different-past structure, apparent continuity can be explained by present cues alone.

5.4 Self-history must concern the system itself, not only the world

The relevant past information must be self-referential in an operational sense. It must encode the system's own prior condition, commitments, tendencies, capability estimates, or internally mediated trajectory.

A memory of what happened in the environment is not enough unless it is functionally tied to the system's own prior states and future policy.

5.5 Temporal separation must be meaningful

The continuity effect must survive meaningful temporal separation, such as:

- phase boundaries,
- reset boundaries,
- interruptions,
- session gaps,
- or cross-episode carry-over.

If the effect only exists as a brief recurrent trace within a single uninterrupted sequence, Level-4 is not yet supported.

5.6 Causal intervention on the continuity pathway

Level-4 requires direct interventions on the self-continuity pathway. At minimum, the evaluation must include one or more of:

- self-history wipe or reset,
- selective degradation or lesion of self-history,
- self-history swap across parallel runs or agents,
- policy–self-history decoupling,
- replacement of integrated self-history with shuffled or world-history-only content.

If performance or behavioural identity does not change under these interventions, the evidence for Level-4 is not sufficient.

5.7 Competitive baselines

The evaluation must compare the candidate system against baselines that can mimic continuity superficially, including:

- reactive or present-only policies,
- Level-3-style target persistence without self-history integration,
- episodic retrieval systems that can recall past events but do not maintain an integrated self-state,
- externally labelled profile systems that receive a trait or identity token from outside rather than constructing continuity internally.

If these baselines can match the candidate on the Level-4 probes, the claim is weakened or rejected.

5.8 Multi-run repeatability

Level-4 attribution must be based on repeated trials:

- multiple seeds,
- multiple self-history trajectories,
- multiple probe points,
- multiple intervention schedules,
- reported variance, not single illustrative runs.

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6. Theoretical Grounding

Level-4 is motivated by convergent ideas across several aligned traditions.

- **Developmental and enactive perspectives:**
  Organised systems are not defined only by current input-output behaviour, but by

continuity of self-maintaining organisation across time.

- **Predictive processing / active inference (conceptual link):**
  Current control can depend on latent estimates shaped by prior self-relevant error, uncertainty, and self-updating history.

- **Autobiographical and self-memory perspectives (conceptual link):**
  Temporal continuity requires more than access to prior episodes; it requires that prior self-relevant history be integrated into present organisation.

- **State-space and control perspectives:**
  When current observation underdetermines the appropriate policy, latent history becomes necessary. In Level-4, part of that latent history is explicitly self-relevant.

These traditions do not establish Level-4 by themselves. They motivate why temporal self-continuity is a meaningful organisational boundary that can be tested without asserting consciousness or full selfhood.

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7. Hypotheses

The evaluation tests the following hypotheses.

H1: History-dependent action selection
Under matched present conditions, the candidate system's behaviour varies systematically as a function of integrated past self-state.

H2: Continuity advantage
On tasks where correct present behaviour depends on prior self-state, the candidate system performs significantly better than present-only, Level-3-only, and episodic-retrieval baselines.

H3: Causal necessity of integrated self-history
If the integrated self-history is wiped, degraded, swapped, or decoupled from policy, behaviour degrades or shifts in the direction predicted by the intervention.

H4: Non-reducibility to episodic recall or external labels
A system with access to isolated episodic memory, world history, or externally supplied identity labels cannot match the candidate's performance and intervention pattern.

H5: Repeatability across runs
The above effects replicate across multiple runs, seeds, self-history trajectories, and probe timings.

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8. Evaluation Methodology

This section defines how evidence for Level-4 should be measured and reported, independent of the specific environment implementation.

8.1 Experimental structure

A Level-4 evaluation should be:

- multi-episode or multi-phase,
- resettable and tightly controlled,
- designed around same-present / different-past probes,
- capable of preserving or disrupting self-history independently of present task cues,
- trajectory-based, since continuity is expressed across time rather than at a single step.

A general Level-4 regime should include at least three components:

1. **History formation phase**
   The system acquires a self-relevant history through prior interaction, adaptation, commitment, or self-updating.

2. **Temporal separation phase**
   A reset, interruption, or neutral interval occurs such that continuity cannot be reduced to immediate stimulus persistence.

3. **Probe phase**
   The system encounters present conditions that are matched across comparison cases, while correct or characteristic behaviour depends on integrated past self-state.

8.2 Primary metrics

At minimum, a Level-4 evaluation should report:

- **Continuity-dependent success rate**
  Fraction of probe trials solved correctly when success depends on prior self-state.

- **History-conditioned policy divergence**
  Degree to which actions diverge across matched present conditions as a function of differing self-history.

- **Intervention sensitivity**
  Change in continuity-dependent performance under wipe, swap, degradation, or decoupling interventions.

- **Self-history transfer effect**
  In swap conditions, the degree to which behaviour shifts with the transferred self-history.

- **Cross-temporal self-state dependence**
  A quantitative measure showing that current behaviour depends on integrated self-history after controlling for present observation and active goal.

8.3 Supporting metrics

Supporting evidence may include:

- latency to continuity-consistent behaviour after a reset or interruption,
- stability of self-relevant behavioural profile across separated phases,
- predictive validity of the internal self-history variable for future policy,
- degradation curve as self-history is progressively lesioned or compressed,
- variance across seeds and self-history schedules.

8.4 Baselines

To avoid false positives, the evaluation should include strong baseline classes.

Baseline 1 — Present-only reactive controller
Policy depends only on current observation and active goal.
Purpose: tests whether present cues are sufficient.

Baseline 2 — Level-3-style goal persistence controller
Maintains internal target information across delay and interference, but no integrated self-history.
Purpose: separates hidden-goal persistence from temporal self-continuity.

Baseline 3 — Episodic retrieval controller
Can retrieve stored past events, but lacks an integrated self-representation that conditions policy.
Purpose: separates memory access from continuity of self-state.

Baseline 4 — External-profile controller
Receives a profile label, trait token, or external identity descriptor at probe time.
Purpose: rules out the explanation that behaviour is driven by externally supplied identity rather than internally maintained self-history.

Baseline 5 — Random or shuffled self-history control
Maintains a memory structure of similar size or activity, but its content is shuffled, mismatched, or unrelated to actual prior self-state.
Purpose: rules out the explanation that any persistent memory trace is enough.

8.5 Intervention protocol

A Level-4 claim requires direct interventions on the continuity-supporting mechanism.

Recommended minimum interventions:

Intervention A — Self-history wipe/reset
The integrated self-history is reset before a probe phase.
Expected Level-4 pattern: continuity-dependent performance falls sharply.

Intervention B — Policy–self-history decoupling
The self-history continues to update, but the policy is forced to ignore it.
Expected pattern: behaviour approaches Level-3 or present-only baseline performance.

Intervention C — Self-history swap
Integrated self-history states are exchanged across matched parallel runs or agents before probe.
Expected pattern: behaviour shifts in the direction of the swapped self-history.

Intervention D — Selective degradation
The self-history is partially lesioned, compressed, or noised rather than fully removed.
Expected pattern: graded decline in continuity-consistent behaviour, not merely all-or-none collapse.

8.6 Reporting standards

The paper should report:

- number of seeds, runs, and probe trials,
- how self-history trajectories were formed,
- how temporal separation was enforced,
- intervention timing and schedule,
- aggregated metrics with variance,
- representative probe trajectories,
- explicit failure cases and null results.

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9. Falsification Criteria

Level-4 is rejected, for a given system and evaluation, if any of the following holds:

- current behaviour under probe conditions is adequately explained by present observation and active goal alone,
- a Level-3-style persistence system matches probe performance without integrated self-history,
- episodic retrieval alone matches the candidate's continuity-dependent behaviour,
- externally supplied profile labels can replace internally maintained self-history without loss,
- wiping, degrading, swapping, or decoupling self-history does not meaningfully alter present behaviour,
- the effect exists only within a single uninterrupted episode and does not survive meaningful temporal separation,

- results do not replicate across controlled reruns.

Negative results remain scientifically informative. They would clarify that the tested mechanism supports memory, persistence, or adaptation of some kind, but not temporal self-continuity in the stricter Level-4 sense.

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10. System design

This section describes the actual implemented Level-4 design used in the accompanying executable notebook. The aim is to test whether present behaviour depends on an integrated representation of prior self-state, rather than only on current observation, active goal, or simple episodic recall. The design is intentionally minimal and should be read as a controlled methodological test rather than as a rich model of selfhood or consciousness.

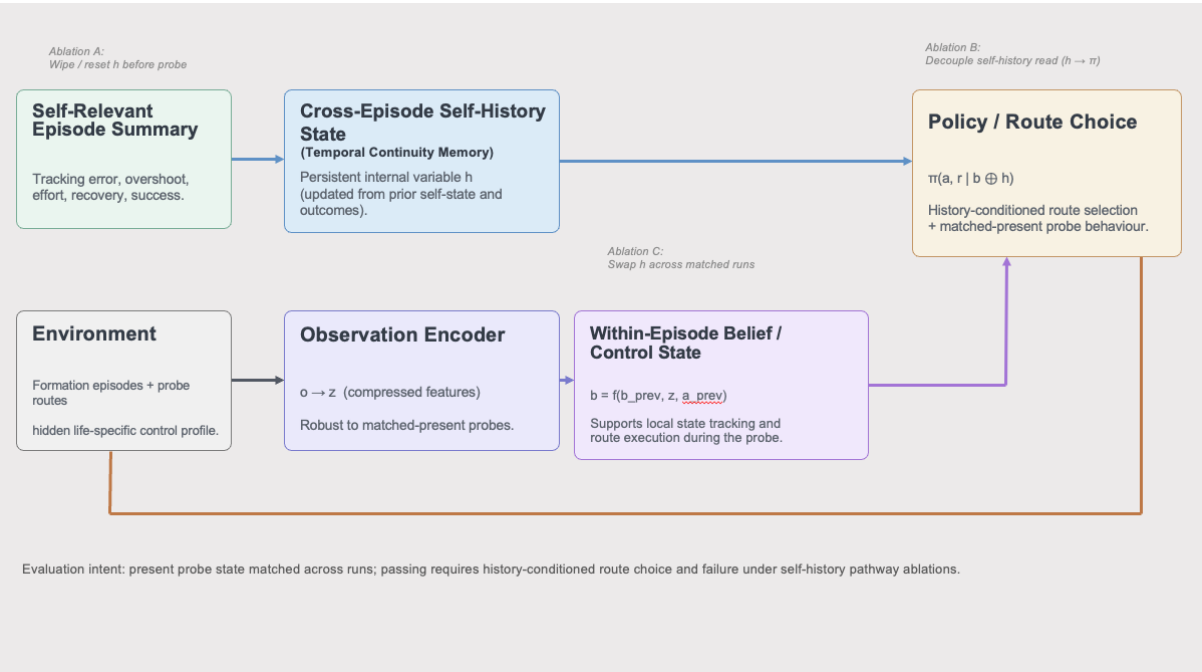

Figure 1. Level-4 architecture for temporal self-continuity under matched-present probe conditions

Each run is organised as a life block with three parts:

1. a **self-history formation block**,
2. a **temporal separation boundary**, and
3. a **matched-present probe block** in which current route choice depends on carried self-history.

Unlike the earlier draft formulation, the implemented system does not use a full continuous-control simulator with explicit velocity dynamics. Instead, it uses a controlled synthetic environment in which hidden life-specific tendencies are sampled, indirectly expressed

through repeated formation episodes, integrated into a self-history state, and then tested through a matched-present route-selection probe.

10.1 Hidden life profile and formation block

For each life block, the system samples a hidden life profile:

$$\ell = (\text{precision_skill}, \text{stability_skill}, \text{efficiency}, \text{drift_phase})$$

where `precision_skill`, `stability_skill`, and `efficiency` are continuous latent properties in $[0, 1]$, and `drift_phase` is used only to generate visually distinct probe trajectories in the illustrative plots. These latent variables are not directly observable by the candidate. Instead, they shape how the system performs across a fixed sequence of five formation episode families:

- corridor
- slalom
- recovery
- mixed
- stop

Each formation episode places different demands on precision, stability, and efficiency. At the end of each episode e, the system produces a self-relevant summary:

$$\sigma_e = [\text{tracking error}, \text{overshoot}, \text{effort}, \text{recovery latency}, \text{success}]$$

These summaries are the basis of temporal self-continuity in the implemented design. They describe how the system itself has been performing across prior episodes, not merely what has happened in the environment.

10.2 Integrated self-history

The candidate does not rely on one remembered event. Instead, the full set of formation summaries is integrated into a persistent cross-episode history state:

$$h = (h_{\text{precision}}, h_{\text{stability}}, h_{\text{efficiency}})$$

where $h_{\text{precision}}$, $h_{\text{stability}}$, and $h_{\text{efficiency}}$ are weighted estimates inferred from the formation block as a whole. The integrated history state also retains the averaged episode-summary variables themselves. Operationally, this means the candidate's present behaviour depends on an accumulated estimate of how it has been across prior episodes, rather than on the final episode alone or on any externally supplied identity token.

A temporal separation is then imposed structurally: formation is completed, the integrated history is fixed, and the subsequent probe is evaluated under matched visible present

conditions. This prevents the probe from functioning as a simple continuation of one long uninterrupted episode.

10.3 Matched-present probe

The probe phase presents the same visible present structure across comparison cases. Each probe belongs to one of several held-out variants, but within each variant the visible start state, target, and route families are fixed. The available routes are:

- precision
- stability
- robust

The defining Level-4 condition is that identical visible present conditions can still yield different correct present choices because different life histories imply different current self-relevant capabilities and liabilities. In other words, the candidate should choose differently now because of its integrated past self-state, not because the probe visibly tells it what to do.

Each route family is defined by different latent demands on precision, stability, and efficiency. The oracle route for a life is computed from the hidden life profile using the route-cost family:

$$c_{\text{precision}} = 1.35(1-p) + 0.60(1-e) + 0.05$$
$$c_{\text{stability}} = 0.62(1-p) + 1.10(1-s) + 0.10$$
$$c_{\text{robust}} = 0.18(1-p) + 0.18(1-s) + 0.72$$

where p, s, and e are the hidden precision, stability, and efficiency values for that life. The implemented candidate uses the same structural cost family, but applies it to the integrated history variables $h_{\text{precision}}, h_{\text{stability}}, h_{\text{efficiency}}$ rather than to the hidden life profile directly.

This makes the probe explicitly continuity-sensitive. Success, time-to-goal, and violation ratio are all computed as functions of route mismatch and demand mismatch, so present behaviour is rewarded only when the system carries forward an adequate integrated self-history into the matched-present choice.

10.4 Candidate, baselines, and interventions

The Level-4 candidate has three essential elements:

1. self-history formation summaries σ_e,
2. an integrated cross-episode self-history state h, and
3. a history-conditioned route-selection rule.

Its present route choice is given by:

$$r = \arg \min_{k \in \{\text{precision,stability,robust}\}} c_k(h_{\text{precision}}, h_{\text{stability}}, h_{\text{efficiency}})$$

The evaluation includes several baselines designed to separate temporal self-continuity from weaker explanations: a present-only reactive controller, an episode-recurrent/no-history controller, a Level-3-style target-only controller, a last-episode episodic controller, and an external profile-token controller. Together these test whether matched-present probe performance can be explained by present-only heuristics, target pursuit, recency, or externally supplied identity information.

The notebook also includes four continuity-pathway interventions:

- **history wipe**, replacing life-specific history with population-prior history,
- **policy–history decoupling**, leaving history present but preventing it from conditioning route choice,
- **history swap**, replacing a life's own history with donor history from another life sharing the same visible probe layout,
- **progressive degradation**, gradually corrupting the integrated history state.

These interventions are central to the Level-4 claim because they test whether present behaviour depends causally on integrated self-history rather than only correlating with it.

10.5 Evaluation metrics and interpretation

The implemented evaluation reports the following primary metrics at the life-block level:

- probe success,
- route match,
- time-to-goal,
- violation ratio,
- policy divergence.

Additional analyses include swap-follow behaviour, degradation curves, route confusion matrices, and representative probe trajectories. Taken together, these measures are intended to show whether current matched-present behaviour tracks integrated cross-episode self-history and whether that dependence fails appropriately under causal intervention.

This implementation is designed to sit strictly above Level-3 and below richer later-level claims. It is above Level-3 because probe behaviour is not explained only by visible target pursuit or short-horizon persistence. It remains below stronger claims about autobiographical narrative, unified self-modelling, or consciousness. The supported claim is narrow: within this controlled synthetic testbed, present behaviour depends on an integrated representation of prior self-state in a causally testable way.

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11. Results

The accompanying executable notebook provides the authoritative operational implementation of the present design and is the source of the multirun results reported below.

11.1 Multirun Evaluation

A multirun evaluation was conducted across 480 life blocks spanning 6 seeds, with the same matched-present probe structure applied across all agent classes. The oracle probe distribution was non-collapsed, comprising 37.3% precision, 26.2% stability, and 36.5% robust probe-optimal routes. The intact Level-4 candidate likewise exhibited a non-degenerate route-selection profile, choosing precision on 236 lives, robust on 124, and stability on 120, indicating that present policy did not collapse to a single route preference under matched visible conditions (Table 1; Figure 1).

Across the full evaluation, the intact candidate_level4 produced the strongest result on the primary Level-4 metrics. It achieved probe success = 0.973, route match = 0.823, and policy divergence = 0.805, exceeding all baselines on the core criterion of history-appropriate route selection under matched present conditions. By comparison, the external-profile baseline achieved route match = 0.517, the present-only reactive controller 0.365, the Level-3-style target-only controller 0.373, the episode-recurrent/no-history controller 0.262, and the episodic last-only controller 0.379 (Table 1; Figure 1).

```
Per-agent overview (mean ± bootstrap CI):
- candidate_level4              success=0.973 [0.956,0.985] | route_match=0.823 [0.790,0.856] | time=88.3 | divergence=0.805
- present_only_reactive         success=0.860 [0.829,0.892] | route_match=0.365 [0.321,0.407] | time=110.4 | divergence=0.000
- episode_recurrent_nohistory   success=0.735 [0.695,0.775] | route_match=0.263 [0.223,0.304] | time=96.7 | divergence=0.000
- level3_target_only            success=0.752 [0.713,0.789] | route_match=0.373 [0.331,0.420] | time=81.2 | divergence=0.000
- episodic_last_only            success=0.825 [0.790,0.858] | route_match=0.379 [0.335,0.425] | time=97.6 | divergence=0.273
- external_profile_token        success=0.860 [0.827,0.891] | route_match=0.517 [0.474,0.562] | time=95.2 | divergence=0.622
- candidate_wipe_h              success=0.740 [0.700,0.781] | route_match=0.373 [0.331,0.420] | time=81.1 | divergence=0.000
- candidate_decouple_h          success=0.729 [0.688,0.767] | route_match=0.373 [0.331,0.420] | time=80.9 | divergence=0.000
- candidate_swap_h              success=0.808 [0.773,0.844] | route_match=0.354 [0.315,0.400] | time=92.4 | divergence=0.619
```

Table 1 — Multi-run summary statistics across agent classes.
Probe success, route match, time-to-goal, and policy divergence for the intact Level-4 candidate, baselines, and ablation conditions.

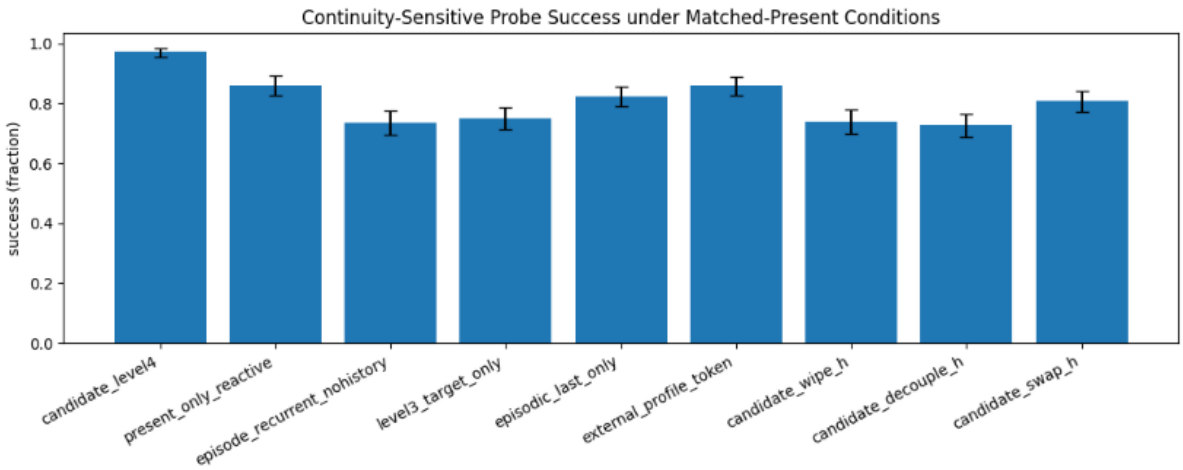


Figure 1 — Continuity-sensitive probe performance across agent classes.
Bar plots showing probe success, route match, and policy divergence under matched-present probe conditions.

11.2 Baseline separation

The baseline pattern is important because Level-4 is not defined by raw task success alone, but by current behaviour depending on an integrated representation of past self-state rather

than simple episodic recall. The framework defines Level-4 exactly in those terms, and distinguishes it from Level-3 target persistence and later broader identity claims.

Several baselines achieved moderate to high probe success without matching the candidate's Level-4 profile. The present-only reactive controller and the external-profile token controller both reached 0.860 success, yet remained well below the candidate on route match (0.365 and 0.517, respectively). The Level-3-style target-only baseline reached 0.752 success but only 0.373 route match, showing that visible target pursuit alone does not recover the history-dependent policy pattern. The episodic last-only baseline improved over purely reactive or target-only control (0.825 success), but still remained far below the intact candidate on route match (0.379), supporting the paper's central distinction between integrated self-history and recency-based episodic access (Table 1; Figure 2).

The candidate confusion matrix was also directionally consistent with Level-4 attribution. It was predominantly diagonal, indicating that route choice tracked the oracle route class across the three probe families, whereas the last-episode-only baseline remained visibly misaligned and biased toward non-optimal route selection (Figure 2). This pattern supports the claim that the candidate's current action policy was conditioned by a more integrated cross-episode representation than the episodic baseline could recover.

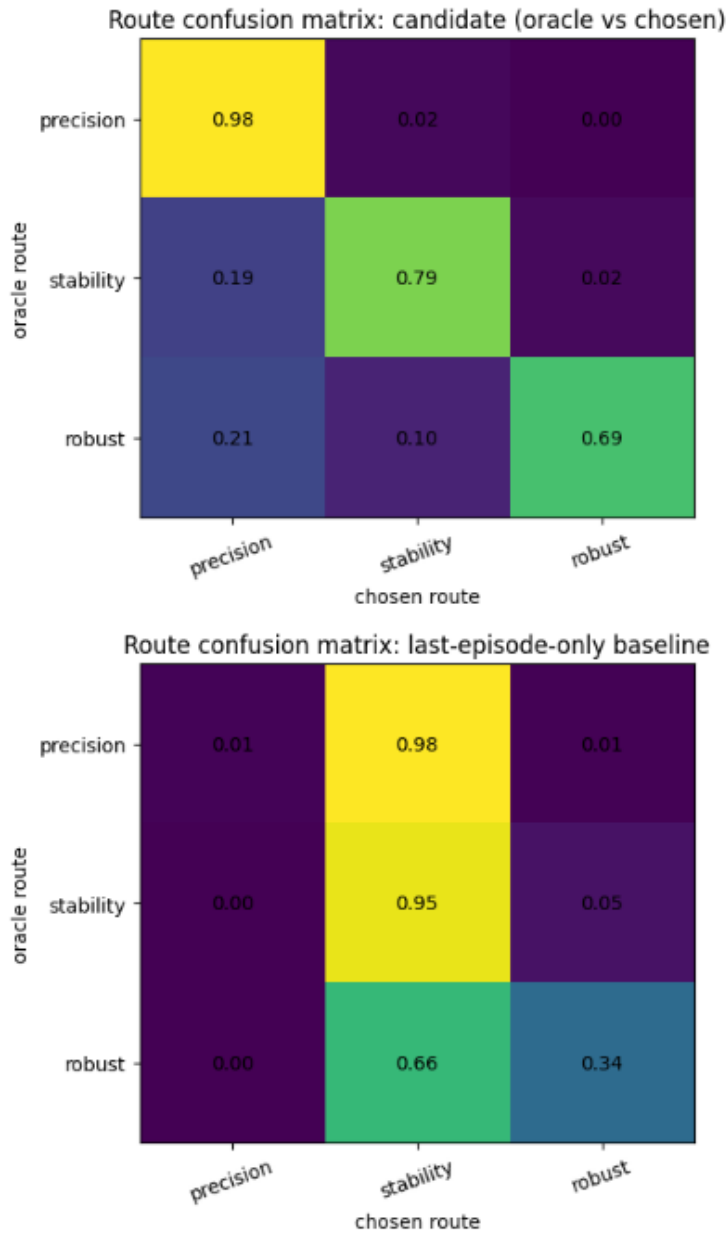


Figure 2 — Oracle-versus-chosen route confusion matrices.
Confusion matrices for the intact Level-4 candidate and the last-episode-only baseline, showing that integrated self-history yields substantially better history-appropriate route selection.

11.3 Causal ablation results

The decisive evidence for Level-4 comes from the continuity-pathway ablations. When the self-history state was wiped immediately prior to the probe (candidate_wipe_h), performance fell to probe success = 0.740 and route match = 0.373. When the self-history state continued to exist but was decoupled from probe policy (candidate_decouple_h), performance fell further to probe success = 0.729 and route match = 0.373. In both cases, the high route-match advantage of the intact candidate disappeared, and the system collapsed toward the no-history / target-only regime (Table 1; Figure 1).

This ablation pattern is the strongest evidence that the relevant cross-episode state was causally necessary rather than cosmetically present. That is also the standard imposed by

the framework: level attribution requires explicit metrics, falsification tests, transparent reporting, and epistemic restraint regarding stronger claims.

11.4 Swap and degradation tests

The self-history swap test provides a stronger intervention on temporal continuity itself. Under swap, the candidate followed its own intact route only 0.404 of the time, while following the donor route with mean 1.000. Moreover, donor and own routes differed on 0.596 of evaluated lives, showing that the swap effect was not vacuous. Under matched present conditions, route choice therefore moved with the transferred self-history rather than remaining fixed by the visible probe state alone (Figure 3).

The degradation sweep produced the same qualitative result. With intact self-history (degradation = 0.00), the candidate showed probe success = 0.977 and route match = 0.823. As corruption increased, both measures declined: at 0.25, success fell to 0.877 and route match to 0.588; at 0.50, success fell to 0.812 and route match to 0.423; and at 1.00, success fell to 0.771 and route match to 0.338. The route-match decline was strong and near-monotone, while probe success showed a clear overall downward trend with only a small stochastic irregularity between intermediate degradation levels (Figure 4).

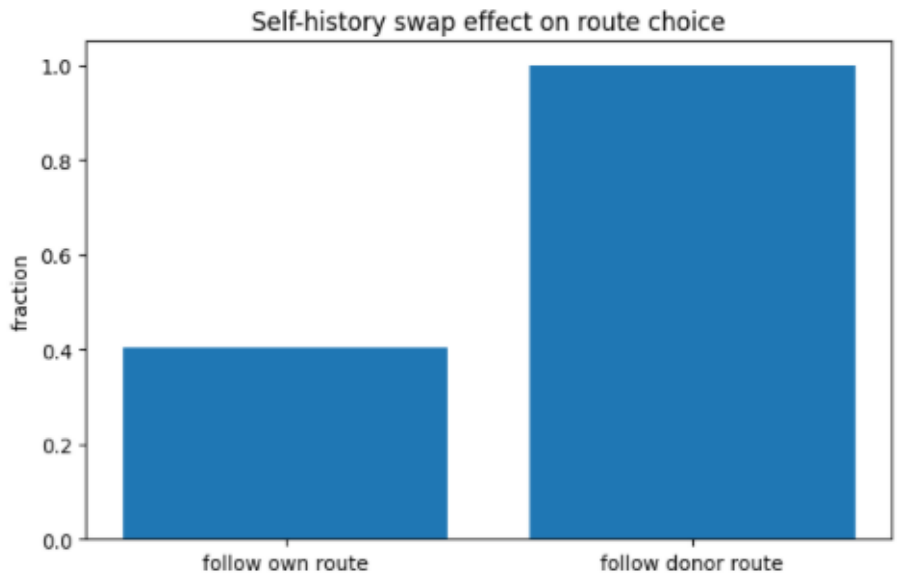


Figure 3 — Self-history swap effect on route choice.
Swap analysis showing that route choice follows transferred donor history under matched-present conditions.

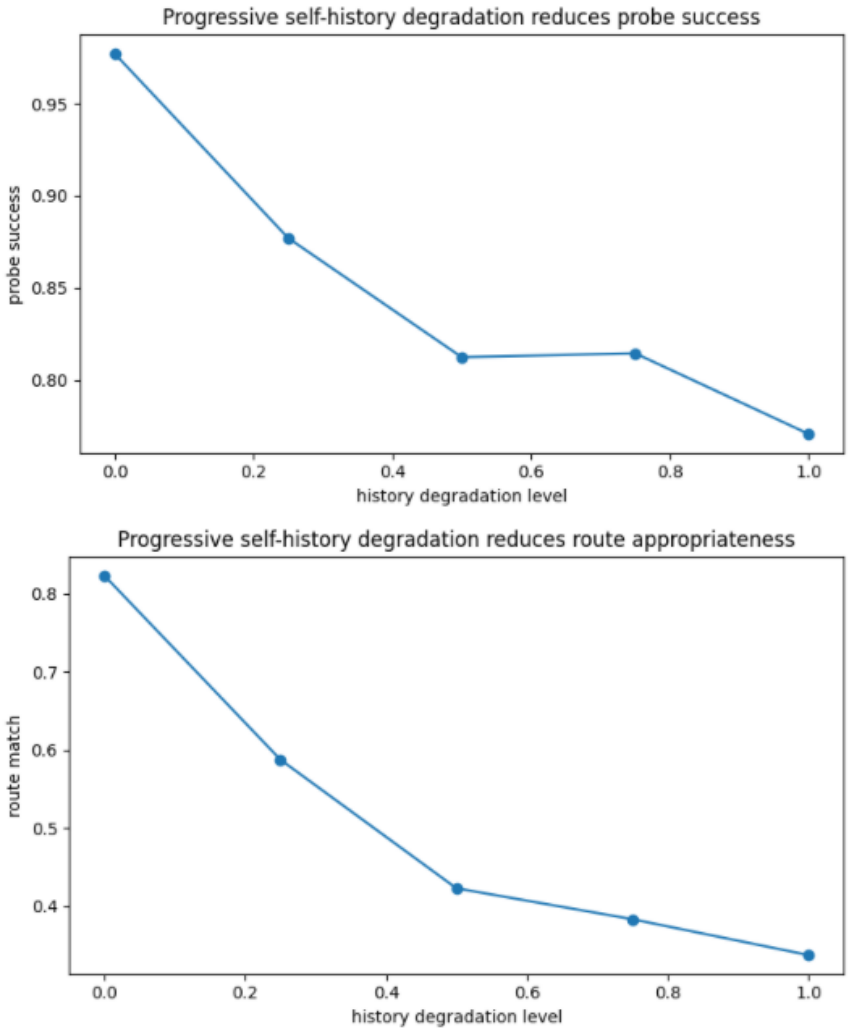
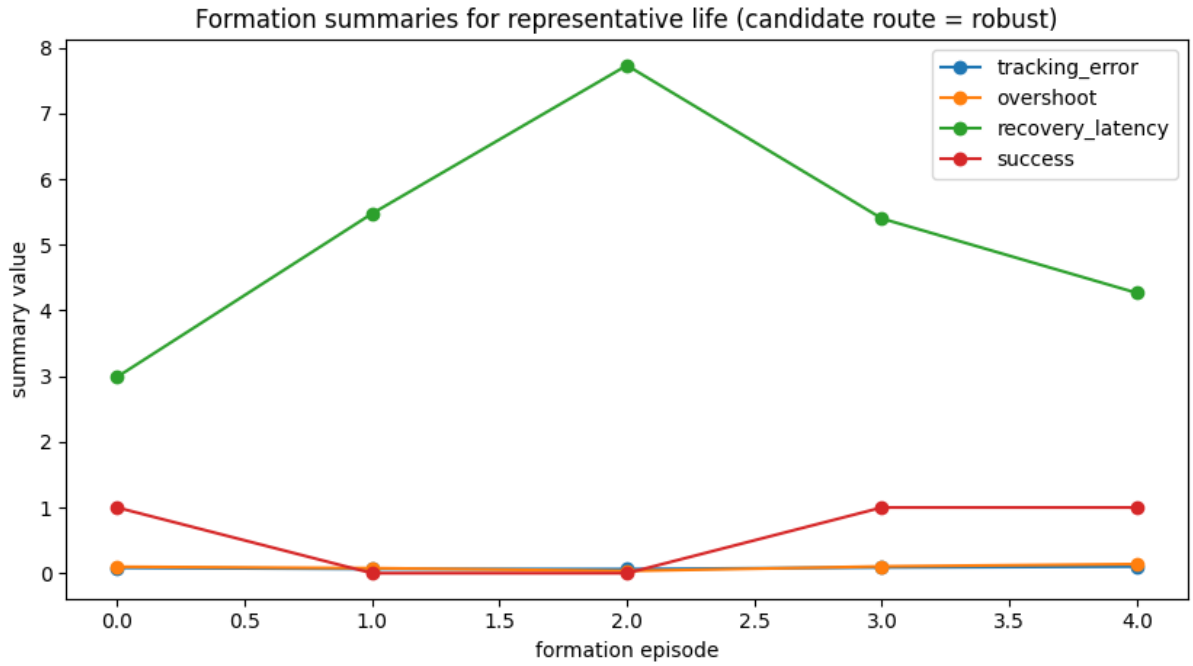
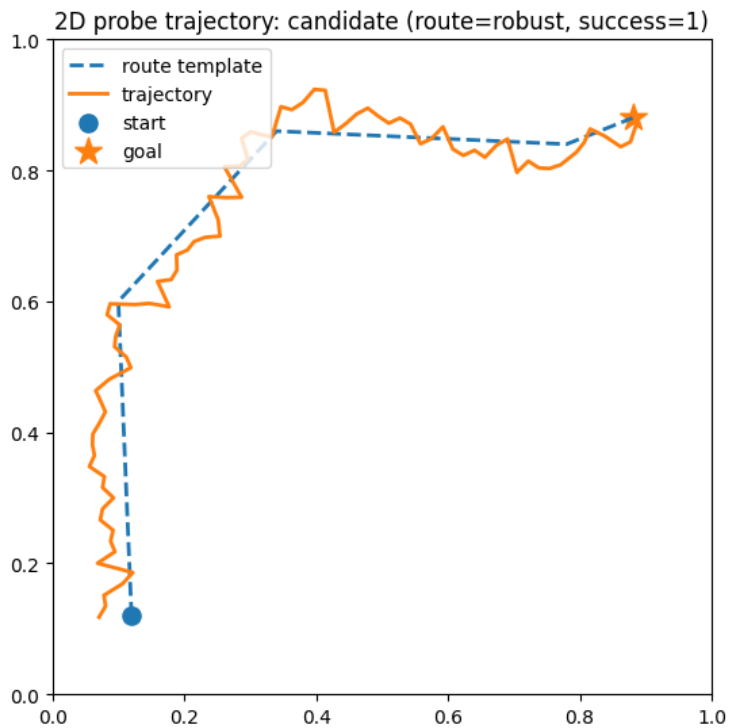


Figure 4 — Progressive degradation of self-history.
Probe success and route-match as a function of self-history corruption level, showing graded degradation rather than all-or-none failure.

11.5 Representative trajectory evidence

To complement the multirun statistics, representative-life traces were examined for formation history and downstream probe behaviour. The notebook generated a formation-history plot for a representative life and paired 2D trajectories for the intact candidate, the Level-3-style target-only baseline, and the history-wipe condition. These qualitative traces provide a useful visual complement to the aggregate metrics because they show the intended causal structure at the level of an individual life block: the candidate's probe route reflects accumulated prior self-state, while target-only and wiped-history conditions revert toward non-Level-4 policy structure (Figure 5).


Formation summaries for representative life (candidate route = robust)
tracking_error
overshoot
recovery_latency
success
summary value
formation episode
0
1
2
3
4
5
6
7
8
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0


2D probe trajectory: candidate (route=robust, success=1)
route template
trajectory
start
goal
0.0
0.2
0.4
0.6
0.8
1.0

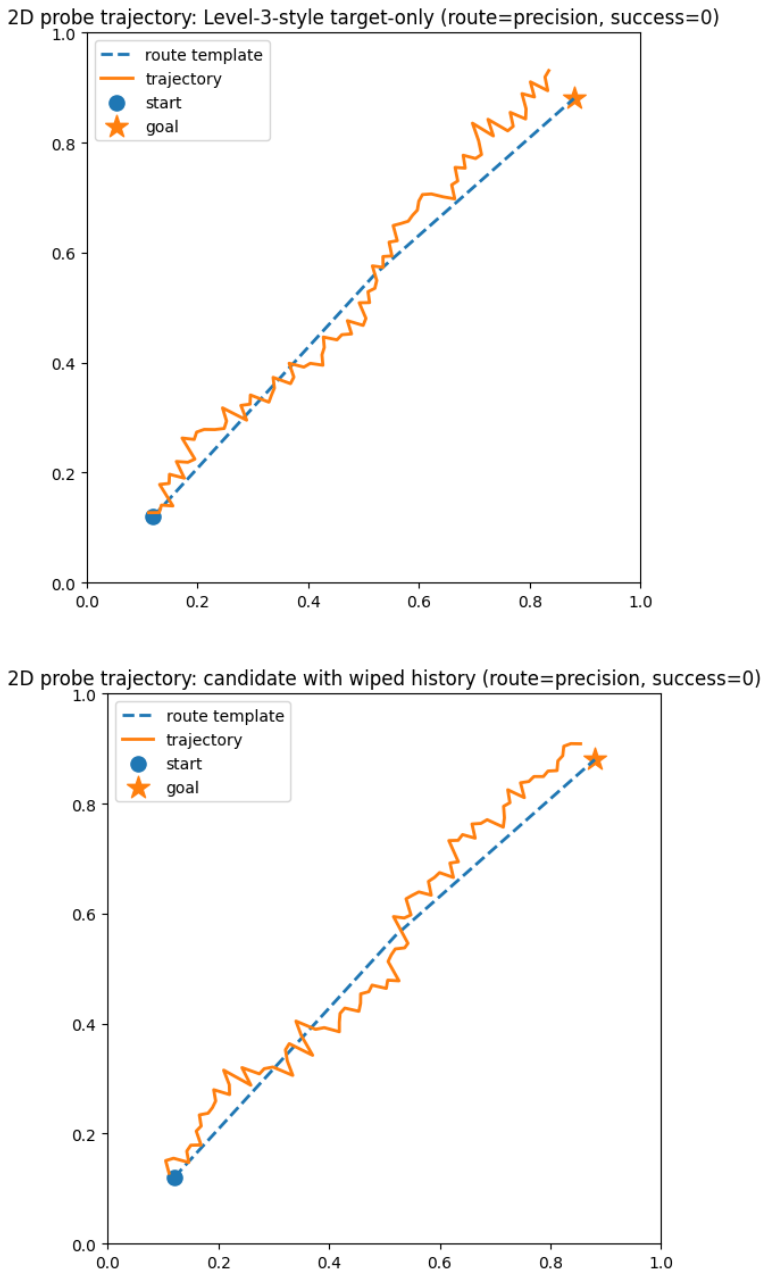


Figure 5 — Representative life-block evidence.
Formation summaries across episodes and example 2D probe trajectories for the intact candidate, target-only baseline, and wiped-history condition.

11.6 Interpretation of empirical structure

Taken together, the results support three bounded conclusions.

First, the Level-4 condition is empirically discriminative in this controlled substrate. Not all systems with memory, target pursuit, or partial task success satisfy the criterion. The decisive difference is whether present matched-probe behaviour tracks an integrated cross-episode self-history and fails appropriately when that pathway is disrupted.

Second, integrated self-history is stronger than simple recency-based episodic access. The last-episode-only baseline shows that recent memory helps, but it does not recover the route-

match or ablation sensitivity of the intact candidate. This matters because the framework defines Level-4 specifically as history-dependent identity rather than simple episodic recall.

Third, the result remains methodological rather than metaphysical. The framework explicitly requires epistemic restraint, prohibits premature consciousness claims, and treats operational indicators as distinct from experience. Your earlier papers also follow that pattern, interpreting successful results as evidence only for the narrow organisational property targeted at that level, not for selfhood or consciousness more broadly.

11.7 Key Limitations

The present implementation should be interpreted in light of three important limitations. First, the candidate uses a designed history-inference pipeline rather than a learned history-integration mechanism. Second, the formation history is analytically generated from the latent life profile, rather than being acquired through prior lived interaction by a trained agent. Third, the temporal separation is structural and minimal: although the probe is separated from the formation block, the current design does not yet test stronger continuity pressures such as decay, interference, interruption, or session-gap persistence.

11.8 Epistemic scope

These results therefore justify a narrow claim only: within the implemented matched-present probe regime, the tested candidate maintains a causally relevant integrated representation of prior self-state, and present behaviour depends on that representation in a way that is not reducible to present-only control, target-only persistence, or last-episode recall. They do not justify stronger claims concerning intelligence, agency, rich autobiographical memory, selfhood in the full philosophical sense, subjective experience, or phenomenal consciousness. That bounded interpretation is exactly in line with the framework and with the style of the earlier level papers, which treat each level as a falsifiable empirical boundary rather than a direct proof of consciousness-related capacities.

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12. Conclusion

This paper defined and evaluated Level-4: Temporal Self-Continuity within the staged Conscious Machines research program. Level-4 requires more than internal state, adaptive regulation, or persistent goal pursuit. It requires that present behaviour depend on an integrated representation of prior self-state, such that under matched present conditions the system's current policy is shaped by who it has been, not only by what it currently observes or what target is presently active. That interpretation is consistent with the broader framework, which defines Level-4 as current behaviour depending on an integrated representation of past self-state and requires level-specific empirical investigation with explicit epistemic restraint.

The reported results support a Level-4-positive interpretation under the operational definition used here. Across multi-run matched-present probe evaluations, the intact Level-4 candidate showed substantially stronger history-appropriate route selection than present-only, episode-

recurrent, Level-3-style target-only, and last-episode episodic baselines. More importantly, the ablation pattern was directionally consistent with causal attribution: wiping the self-history state or decoupling it from policy caused a marked collapse toward no-history behaviour, while swapping the self-history state shifted route choice with the transferred history and progressive degradation of that state produced an overall graded decline in performance. In the bounded methodological sense relevant to this research program, these results support the claim that the implemented system satisfies the Level-4 criterion of temporal self-continuity.

This remains a deliberately narrow conclusion. As in the earlier level papers, the present result does not imply intelligence, agency, rich autobiographical memory, selfhood in the full philosophical sense, subjective experience, or phenomenal consciousness. The program's own evidential standard is that each level supports only its local operational definition and nothing stronger; operational indicators remain distinct from phenomenology, and negative or limited findings remain scientifically meaningful.

The significance of the present work is therefore methodological rather than metaphysical. It shows that temporal self-continuity can be translated into a concrete, falsifiable, reproducible experimental design rather than left as a purely intuitive or narrative notion. In that respect, Level-4 extends the ladder in the intended way: from internal mediation, to adaptive regulation, to persistent target maintenance, and now to a minimal form of history-dependent self-continuity that is experimentally isolable and ablation-testable. This is the kind of cumulative, level-specific progress the program framework was designed to support.

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13. Next steps

The immediate next step is to **stress-test the Level-4 result**, not broaden the claim. The present study functions as a minimal engineered existence proof of temporal self-continuity under matched-present probe conditions. The right follow-on work is therefore to test whether the same causal pattern survives under harder and more varied evaluation settings.

Three near-term directions are most important.

First, the probe regime should be broadened. Future work should test additional held-out probe layouts, longer temporal gaps between formation and probe, and more varied self-history-relevant interference. This would help determine whether the present effect generalises beyond the current task family.

Second, the self-history formation block should be made more demanding. More diverse prior episodes, broader disturbance families, and less regular training structure would place stronger pressure on the claim that the system carries forward an integrated representation of prior self-state, rather than exploiting narrow regularities in one environment.

Third, the baselines and ablations should be strengthened further. In particular, future work should compare against richer retrieval-based baselines and more adversarial continuity-pathway interventions, so that the distinction between integrated self-history and simple memory access is tested more severely.

Beyond that, the natural programmatic continuation is **Level-5**, but only under the same evidential discipline used here. The contribution of the present paper is not that it resolves synthetic selfhood or consciousness, but that it makes one further organisational boundary experimentally tractable.